

Practical Week – 2

Task : 1

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>

#define MAX_INPUT 100

int main()
{
    char input[MAX_INPUT];
    char *args[MAX_INPUT];
    int i = 0;
    pid_t pid;
    int status;

    printf("Enter a Linux command: ");
    fgets(input, sizeof(input), stdin);

    // Remove newline
    input[strcspn(input, "\n")] = '\0';

    // Convert input into arguments
    char *token = strtok(input, " ");

    while (token != NULL)
    {
        args[i++] = token;
        token = strtok(NULL, " ");
    }
    args[i] = NULL;

    // Create child process
    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        // Child process
        printf("\nChild Process\n");
        printf("Parent PID : %d\n", getppid());
        printf("Child PID : %d\n", getpid());

        printf("\nExecuting command: %s\n", args[0]);

        // Execute command
        execvp(args[0], args);

        // Only reached if execvp fails
        perror("execvp failed");
        exit(1);
    }
    else
    {
        // Parent process
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        // Wait for child
        wait(&status);

        printf("\nChild process completed.\n");
    }

    return 0;
}
```

Output :

```
anuan@anushka:~$ nano command_executor.c
anuan@anushka:~$ gcc command_executor.c -o command_executor
anuan@anushka:~$ ./command_executor
Enter a Linux command: ls

Parent Process
Parent PID : 492
Child PID : 493

Child Process
Parent PID : 492
Child PID : 493

Executing command: ls
File1.txt      command_executor  keyboard_input      parser_quotes.c
File2.txt      command_executor.c keyboard_input.c     process
LICENSE        docs              main_loop           process.c
Makefile       file_copy         main_loop.c         process.c.save
OSSP           file_copy.c       obj                 scripts
README.md      fork_exec         parser_double_quotes src
assets         fork_exec.c       parser_double_quotes.c tests
bin            include          parser_quotes

Child process completed.
anuan@anushka:~$
```

Task : 2

Code:-

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main()
{
    printf("=====\n");
    printf("    STRACE SYSTEM CALL TRACING\n");
    printf("=====\n");

    printf("Creating sample.txt...\n");
    system("echo 'Hello, this is a sample file.' > sample.txt");

    printf("\nDisplaying sample.txt using cat:\n");
    system("cat sample.txt");

    printf("\nTracing system calls generated by cat...\n");
    system("strace -o trace.txt cat sample.txt");

    printf("\nSystem call trace saved in trace.txt\n");
    printf("\nDisplaying the trace:\n");
    system("cat trace.txt");

    return 0;
}

```

Output:-

```

anuan@anushka:~$ nano command_executor.c
anuan@anushka:~$ gcc command_executor.c -o command_executor
anuan@anushka:~$ ./command_executor
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Parent Process
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OSSP       file_copy.c          obj             scripts
README.md  fork_exec            parser_double_quotes  src
assets     fork_exec.c          parser_double_quotes.c  tests
bin         include              parser_quotes

Child process completed.
anuan@anushka:~$
anuan@anushka:~$ uname -a
Linux Anushka 6.10.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 19 07:09:16 UTC 2026 x86_64 GNU/Linux
anuan@anushka:~$ lscpu | head -20
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 16
On-line CPU(s) list:   0-15
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i7-13620H
CPU family:             166
Model:                 146
Thread(s) per core:     2
Core(s) per socket:     8
Socket(s):              1
Stepping:               2
BogoMIPS:               5836.80
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush
                        mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl
                        VT-x
Virtualization:         Microsoft
Hypervisor vendor:      Microsoft
anuan@anushka:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root         1      0  0 07:00 ?        00:00:00 /sbin/init
root         2      0  0 07:00 ?        00:00:00 /init
root         6      0  0 07:00 ?        00:00:00 /usr/lib/systemd/systemd --switched-root --system --deserialize 22
root        46      1  0 07:00 ?        00:00:00 /usr/lib/systemd/systemd-journald
root        88      1  0 07:00 ?        00:00:00 /usr/lib/systemd/systemd-udevd
message+    109     1  0 07:00 ?        00:00:00 /usr/bin/dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation
root       319    316  0 07:01 pts/0    00:00:00 -bash
root       570      2  0 07:01 ?        00:00:00 /init
root       625    573  0 07:02 pts/2    00:00:00 ps -ef
root       626    319  0 07:02 pts/0    00:00:00 ps -ef
anuan@anushka:~$ top -b -n 1 | head -20
top - 07:02:44 up 2 min, 1 user, load average: 0.00, 0.01, 0.00
Tasks: 29 total, 1 running, 28 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.8 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7892.0 total,  6320.8 free,  601.5 used,  969.7 buff/cache
MiB Swap: 2048.0 total,  2048.0 free,  0.0 used,  7290.5 avail Mem

   PID USER      V#  NI   OPA  RES    SHR   S#O  CWN  %PRC COMMAND
  1 root        0    0  24160 12100  5 0 0  2  0:00.11 /init
  2 root        0  24316 15388 11580  5 0 0  0.2 0:00.02 /sbin/init
  6 root        0  23532  6240  4224  5 0 0  0.1 0:00.00 /init
 319 root        0  18212  3520  2920  5 0 0  0.0 0:00.00 -bash
 625 root        0    0  9100  3280  2884  6 0 0  0.0 0:00.00 ps

```